

ACX Platform (HMS): Hospitality Management System User Manual

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Abstract *The ACX Platform Hospitality Management System (HMS) is the Inspironics web platform through which hotel administrators and duty managers run a smart-building property: bookings and stays, room occupancy and housekeeping condition, guest and facility services, planned maintenance, job orders, Caleido device alerts and operational reporting. This manual describes the platform as a user encounters it: the role and module-permission model, the navigation surface, the recurring workflows book, allocate, check in, serve, check out the dashboard, and the safeguards that constrain every one of them. It documents behaviour as implemented in the baseline build (API v0.3.0, 92-table schema); capabilities the baseline leaves unconnected are marked as such rather than described as though delivered.*

Index Terms *hospitality management, property management, role-based access control, occupancy, stay lifecycle, housekeeping, service requests, smart building, IoT alerts.*

I. INTRODUCTION

ACX HMS exists because a smart hotel generates two kinds of truth that must be read together: the commercial truth of who is staying in which room, and the physical truth of what the building is reporting about that room. The platform places bookings, occupancy, housekeeping condition, services and device alerts on one surface, organised by the facility hierarchy of the property.

The manual is written for the roles defined in III: administrators who configure the property, duty managers who run the shift, and through the permission matrix the front-desk, housekeeping and technical staff whose work the web platform records. It is one of three companion documents; the Quick Reference [2] condenses lookup tables and the Troubleshooting Guide [3] covers fault diagnosis.

Two framings are load-bearing and are repeated here because they govern every screen in the product.

ACX HMS is not a payment gateway, an accounting ledger, a channel manager, a device-firmware console, a messaging service or a guest-facing application. Guests, staff handsets and back-end services have their own platforms and cannot sign in to HMS Web.

ACX HMS is the web platform that holds the property structure, the stays and room allocations, the room status and housekeeping conditions, the service and maintenance workload, and the device alerts of one facility, and that enforces on the server, on every request which role may read or change each of them.

II. SYSTEM SCOPE AND CAPABILITIES

A. Architecture and Boundaries

The platform is three tiers. A React and TypeScript single-page client, served by Vite, talks to a FastAPI service under /api/v1; the service persists to one PostgreSQL database (hms_db) whose 92 application tables are created by a single Alembic migration. There is no cache, queue or scheduler tier: every figure on screen is read from the database at request time. Table I states what each part holds.

TABLE I
COMPONENTS AND WHAT EACH HOLDS

Component	Holds	Relationship
Web client	Presentation, session token, theme	Never authoritative for access
HMS API	Business rules, state transitions, authorization	Sole writer of the database
PostgreSQL	Property, stays, rooms, services, devices, alerts, audit columns	System of record
Caleido devices	Telemetry, alerts, incidents (as data)	Read and acknowledged; not commanded
Staff / guest apps	Mobile workflows	Separate platforms; refused by HMS Web

B. Modules

Eighteen permission modules divide the platform: dashboard, occupancy, bookings, service_tracking, service_planning, facility_management, user_roles, service_setup, employees, job_order, offers, events, caleido_network, firmware_management, reports, tickets, holidays and default_key. Each module declares whether read and write are applicable; dashboard and reports are read-only by construction. A module is the unit a role is granted, and the same grant governs both the screen and the API behind it.

C. Facility Hierarchy

Every room is placed in the hierarchy Organisation Facility Property Building Floor Room. Buildings and floors are projections of the property chain rather than separate records, which is why every building and floor picker in the product is filled from the same source. A room (an *amenity*) carries a type whose category is room, restaurant or others; guest rooms and non-guest spaces are shown on separate tabs throughout. The baseline operates one facility per database.

III. ROLES AND ACCESS MODEL

Access combines a *role*, which decides the modules a user may read or write, with the *platform* the role belongs to. Only roles of type admin or manager belong to HMS Web; staff, guest and system roles belong to the mobile, guest-mobile and service platforms and are refused at sign-in with a message naming their platform. Table II summarises the seeded roles.

TABLE II
SEEDED ROLES AND PRIMARY USE

Role	Type	Primary use
Administrator	admin	All 18 modules, read and write; property configuration
Duty Manager	manager	Shift operation: 13 modules, 10 writable; no facility or role administration
Front Desk	staff	Bookings, occupancy, tickets, default key (mobile platform)
Housekeeping	staff	Room condition, service tracking and planning (mobile)
Technician	staff	Devices, job orders, service tracking (mobile)
Guest	guest	Service requests from the guest application
System	system_user	Service integrations; no interactive sign-in

Administrators may define further roles and edit the read/write matrix of any role, but cannot invent modules, cannot grant write where a module has no write, and cannot grant write without read.

A. Sessions

Sign-in exchanges a username and password for a signed access token valid for 60 minutes by default. The username is matched exactly and case-sensitively; an e-mail address is not a username. There is no refresh token: when the token expires the next request is refused and the client returns to the login page. Roles and grants are re-read on every request, so a permission change takes effect on the user's next action without signing out.

B. Client and Server Enforcement

The client hides sidebar entries and disables buttons for modules the user cannot read or write. That is a convenience only: the API evaluates the same grant on every request and is the authority. A hidden button and a refused request are therefore always consistent, and a request that bypasses the client receives the same answer.

IV. INTERFACE AND NAVIGATION

A. Route Set

The client implements the routes in Table III. Each route requires read on its module; a route the user cannot read shows an *Access denied* page naming the missing module, and unknown paths show *Not Found*.

TABLE III
CLIENT ROUTE SET

Group	Routes
Entry	/login, /HMS (dashboard; /, /hms, /dashboard redirect)
Operations	/occupancy, /bookings, /tickets
Services	/services/tracking, /services/planning
Config and Setup	/config/facility, /config/user-roles, /config/services-setup, /config/employees, /config/job-order, /config/limit-alert
Manage and Stay	/offers, /holidays, /events
Reports	/reports/{key} (nine report tabs)
Keys	/key-settings

The sidebar groups these as HMS, Occupancy, Stay, Services, Config and Setup, Manage, Reports, Tickets and Default Key Settings; a group disappears when none of its entries is permitted. The header carries the sidebar toggle, the notifications bell, the current user, a light/dark theme switch and Logout.

B. The Dashboard

/HMS opens on eight KPI tiles device alerts, active value alerts, incidents, rooms in house, stays in house, service requests, energy consumed and activities followed by average energy consumption, the occupancy donut, the *Caleido At Work* rings (smart rooms online, service request status, rooms for check-out, pending bookings), the alerts panel with Service and Caleido views over today, week or month, recent activity, and the building floor room drill-down. Selecting a donut slice opens the occupancy list filtered to that status.

A tile that cannot be computed says why *No access*, *Session expired* or *Unavailable* rather than showing zero. A zero on this dashboard is a measured zero.

C. The Occupancy Dashboard

Occupancy is where most shift work happens. Guest and Non Guest tabs list rooms filtered by building, floor and housekeeping condition in one scrolling list, with room, type, building, floor, guest, check-in, check-out, a *Generate* action, status, reassign and invoice columns. Opening a room shows its details, environment and utilities (power, energy, temperature, air quality) and its service requests.

D. Status Signalling

The interface marks what it shows rather than leaving the reader to infer it:

room status is colour-coded Available green, Occupied red, Allotted yellow, Unavailable blue, unknown grey; the condition recorded as Dirty is displayed as **Need Maintenance**; a guest still in house after the expected check-out carries a red **Overdue** tag; device health shows only the two states the devices report, Active and Inactive; a capability not yet connected says so (for example *Bulk upload is not connected yet*) instead of failing silently.

V. CORE WORKFLOWS

Fig. 1 shows the loop the platform is built around.

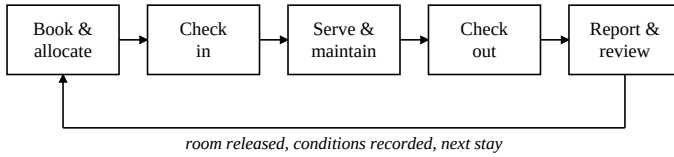


Fig. 1. The recurring HMS operating loop. A room re-enters the loop only when its stay has checked out or been cancelled, and its housekeeping condition travels independently of its status.

A. Creating a Booking

Add New Booking on /bookings creates the guest record and then the stay. A stay carries an internal reference of the form STY-YYYY-NNNN, the booking user, the number of rooms and guests, the expected check-in and check-out times, GST, source (ikanos or porta) and document-approval status. The expected check-out must be later than the expected check-in. Only Available rooms can be allocated, a room can be held by one live stay at a time, and every allocated room becomes **Allotted**. Package and sub-package (Room Only, Breakfast Included, Half Board, Full Board) are recorded against the allocation.

B. Check-In

Check In requires at least one allocated room and a stay that is neither checked in, checked out nor cancelled. It stamps the actual check-in time, moves the stay to active and the rooms to **Occupied**. Guest identity documents are approved separately through Documents Approval.

C. Check-Out

From the Occupancy list, a room with an in-house stay offers **Check-Out** in the Generate column. The action always asks first *Check out Room {n}? This action will release the room.* and names the guest. Confirming stamps the actual check-out time and the initiating user, moves the stay to checked out and every room of the stay to **Available**. The dialog stays open if the server refuses, so the reason is read before anything is retried. Check-out does not change housekeeping conditions: marking a room for cleaning is a separate, deliberate act on the Status cell.

D. Extend, Reassign and Cancel

Extend Checkout Date & Time accepts only a later time than the current expected check-out. *Reassign* moves a stay to another free room: the old room becomes Available and the new one Occupied or Allotted according to whether the guest is in house. *Cancel* is a status change, never a deletion; it releases the rooms and is refused for a checked-in stay, which must be checked out instead. Fig. 2 summarises the lifecycle.

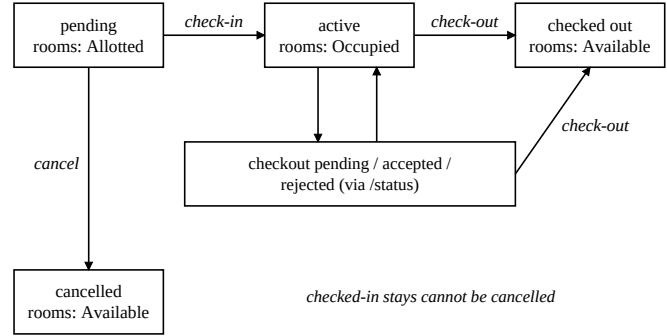


Fig. 2. Stay lifecycle and the room status each state implies. checked out and cancelled are reached only through their own actions, never by setting a status directly.

E. Services, Tickets and Planning

Seven service types Room Service, Travel Desk, Business Center, Food Order, Facility Maintenance, Health & Fitness and Sanitation Maintenance are configured in Services Setup and tracked on Services Tracking and Tickets. A request (SR-YYYY-NNNN) moves Pending Assigned Partially completed Completed, or Canceled; assigning a pending request makes it Assigned, completion stamps the completion time, and a completed request cannot be cancelled. Services Planning schedules recurring work scheduled services, planned maintenance and disinfection against rooms and staff on a weekly rule. Job Orders (JO-YYYY-NNNN) record installation, replacement and troubleshooting work on rooms and devices.

F. Alerts and Limits

Caleido devices (hub, air quality, Mikos and lock) raise alerts of sixteen types, from BatteryLow to LockOffline, at warning or critical severity. An incident moves Unread Read Assigned Resolved and every transition appends a history row; leaving Resolved records a reopening. Limit Config Alert sets the low and high thresholds that raise value alerts; low must be below high.

G. Reports

Nine reports occupancy, employee, room status, booking, ticket, housekeeping, sanitization, alert and energy are generated on screen and exported to Excel. Each is addressed as /reports/{key}.

VI. ADMINISTRATION AND SAFEGUARDS

Administration covers facility setup, amenity types, room amenities, packages, rooms, user roles and the module matrix, service catalogues, employees, departments and job functions, limit configurations, offers, holidays and events. Staff are deactivated with a termination date and may be reactivated; they are never deleted, so every record they touched keeps its author.

Six safeguards are worth stating plainly, because they explain most of the platform's refusals:

authorization is evaluated by the server on every request from the database, not from the token;
HMS Web admits only admin and manager roles; other platforms' accounts are refused by name;
a room held by a live stay cannot be set Available or Unavailable, so occupancy cannot be overwritten by hand;
check-out and cancellation are separate actions with their own preconditions, never a status edit;
records are soft-deleted (status 0), preserving history and references;
the service refuses to sign tokens outside development while the default secret is configured.

VII. GOOD PRACTICE AND KNOWN LIMITATIONS

A. Good Practice

Allocate rooms at booking time. Check-in is refused until a room is allocated.
Read the dialog before confirming check-out; the room is released immediately.
Record housekeeping conditions after check-out the status change does not do it for you.
Grant the narrowest role that does the job; the Duty Manager deliberately lacks facility and role administration.
Treat *No access* on a tile as information about the role, not a fault.
Prefer deactivating staff to editing their records away.

B. Known Limitations of This Baseline

This manual documents implemented behaviour. Several capabilities are present as data or interface but are deliberately unconnected in the baseline: invoice generation (no tariff, price or tax rate exists, so the Invoice column is disabled), payments, bulk booking upload, default key management, PDF export, and delivery of notifications by e-mail, SMS or push. Room status and the stay graph are two sources of occupancy truth that the baseline does not reconcile, so the occupancy donut and *Rooms in house* can differ. The dashboard KPI table is not keyed by facility, which confines the baseline to one facility. Users should read figures depending on these items as provisional; the manual will be reissued against the delivered build.

VIII. CONCLUSION

ACX HMS makes the operation of a smart property structural rather than procedural: rooms change state only through bookings, check-ins and check-outs, every refusal is decided by the server against the user's role, and the dashboard shows why a figure is missing rather than inventing one. For a user, the practical consequence is that the platform will sometimes refuse a hidden module, a blocked check-in, a room that will not be released. Each refusal is a rule stated in this manual. Used as intended, HMS turns a building full of rooms and sensors into one shift that can be run from a single screen.

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